

Understanding My RAG Project (Student-Friendly Guide)

Big Picture

User asks a question. FAISS finds relevant information from the PDF. Gemini reads that information and generates a human-friendly answer.

What is a Chunk?

A chunk is a small piece of a PDF. Splitting documents into chunks makes searching efficient.

What Does FAISS Do?

FAISS acts like a librarian. It searches thousands of chunks and retrieves the most relevant ones.

What Does Gemini Do?

Gemini acts like a teacher. It reads retrieved chunks, understands them, combines information, summarizes, explains, and answers questions.

Does Gemini Search the PDF?

No. FAISS searches the PDF chunks. Gemini only receives the selected context.

What is Context?

Context is the relevant information retrieved from the PDF by FAISS. Gemini uses this context to answer the question.

How Does Gemini Return Answers?

The app sends a prompt containing the question and context. Gemini reads the prompt and generates a response.

Can Gemini Only Rephrase?

No. Gemini can rephrase, merge information, summarize, explain concepts, and answer questions using the context.

Why Not Send the Entire PDF?

Sending the entire PDF is slow, costly, and may exceed token limits. FAISS first narrows down the relevant information.

Best Analogy

PDF = Textbook, Chunk = Small section, FAISS = Librarian, Gemini = Teacher, User = Student.

One-Line Summary

FAISS finds the relevant information. Gemini understands the information and explains it.